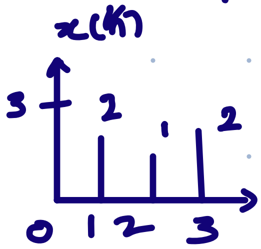
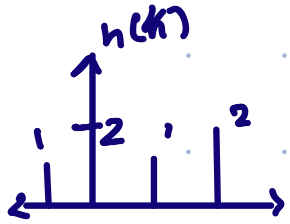
2022-23

	3	2	1	2
1	3	2	1	2
2	6	4	2	4
1	3	2	1	2
2	6	4	2	4

Q1. $x(n) = \{\underset{\uparrow}{\overset{0}{3}}, \overset{1}{2}, \overset{2}{1}, \overset{3}{2}\}$ $h(n) = \{1, \underset{\uparrow}{\overset{0}{2}}, \overset{1}{1}, \overset{2}{2}\}$





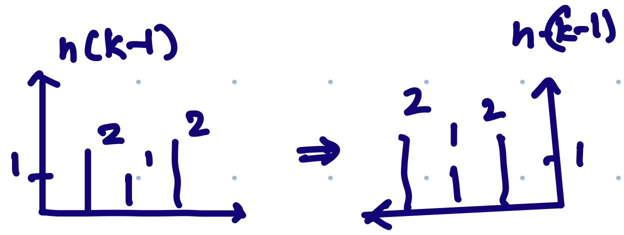
$0 \rightarrow 3$
$-1 \rightarrow 2$

$-1 \rightarrow 5$
↳ n

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(-k+n)$$

n = -1

$$y(-1) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(-k-1)$$

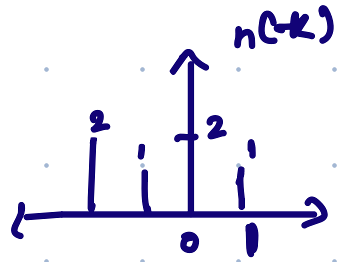


$= 3 \times 1 = 3$

n=0

$y(0) = \sum x(k)\, h(-k)$

$= 3\times 2 + 2 \times 1 = 8$

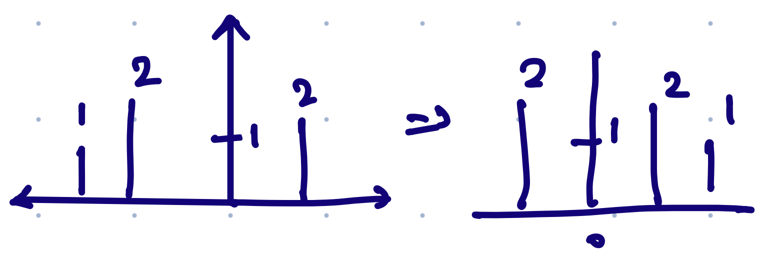


n=1

$y(1) = \sum x(k)\, h(-k+1)$

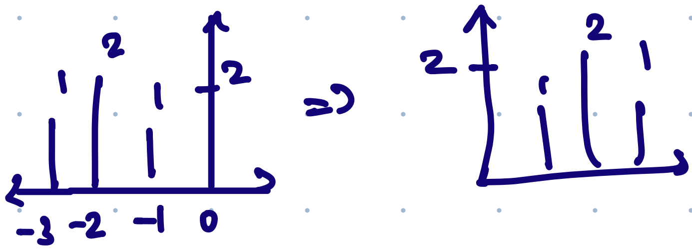
$= 3 \times 1 + 2 \times 2 + 1 \times 1$

$= 8$



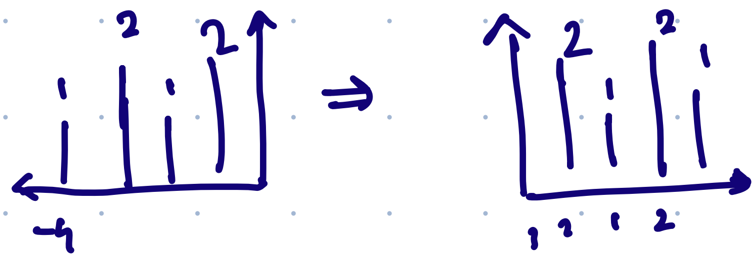
n=2

$$y(2) = \sum x(k)\,h(-k+2)$$
$$= 3\times2 + 2\times1 + 1\times2 + 2\times1$$
$$= 6+2+2+2 = 12$$



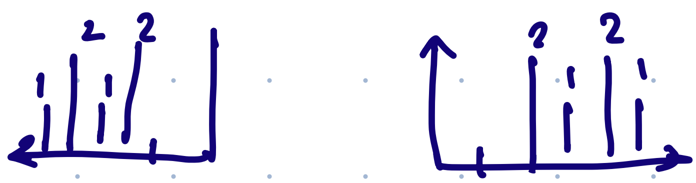
n=3

$$y(3) = \sum x(k)\,h(-k+3)$$
$$= 2\times2 + 1\times1 + 2\times2$$
$$= 9$$



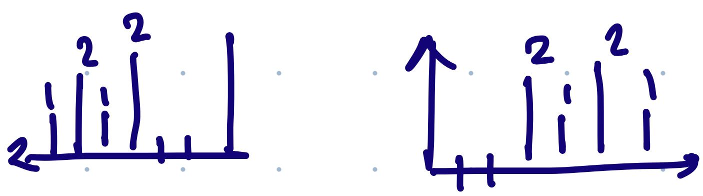
n=4

$$y(4) = \sum x(k)\,h(-k+4)$$
$$= 1\times2 + 1\times2$$
$$= 4$$



n=5

$$y(5) = 2\times2$$
$$= 4$$



$$y(n) = \{3, \underset{\uparrow}{8}, 8, 12, 9, 4, 4\}$$

b) Circular convolution

$x(n) = \{\underset{\uparrow}{-1}, 1, 2, -2\}$ $\quad h(n) = \{0.5, \underset{\uparrow}{1}, -1, 2, 0.75\}$

$0 \rightarrow 3$
$-1 \rightarrow 3$
$\overline{-1 \rightarrow 6}$

$N_1 + N_2 - 1 = 4 + 5 - 1 = 8 \swarrow$ 0 to 7

0 $\quad$ -1 $\quad$ 1 $\quad$ 2 $\quad$ -2 $\qquad$ x(m)

0.5 $\quad$ 1 $\quad$ -1 $\quad$ 2 $\quad$ 0.75

pad

0 $\quad$ -1 $\quad$ 1 $\quad$ 2 $\quad$ -2 $\quad$ 0 $\quad$ 0 $\quad$ 0

0.5 $\quad$ 1 $\quad$ -1 $\quad$ 2 $\quad$ 0.75 $\quad$ 0 $\quad$ 0 $\quad$ 0

(n-m)

-1 to 6

	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7
$x_1(m)$							0	-1	1	2	-2	0	0	0	
$x_2(m)$							0.5	1	-1	2	0.75	0	0	0	
$x_2(-1-m)$	0	0	0	0.75	2	-1	1	0.5							
$x_2(-m)$		0	0	0	0.75	2	-1	1	0.5						
$x_2(1-m)$			0	0	0	0.75	2	-1	1	0.5					
$x_2(2-m)$				0	0	0	0.75	2	-1	1	0.5				
$x(3-m)$					0	0	0	0.75	2	-1	1	0.5			
$x(4-m)$						0	0	0	0.75	2	-1	1	0.5		
$x(5-m)$							0	0	0	0.75	2	-1	1	0.5	
$x(6-m)$								0	0	0	0.75	2	-1	1	0.5

$$x_3(-1) = \sum_{k=-\infty}^{\infty} x_1(m) \cdot x_2(-1-m) = -0.5$$

$$x_3(0) = \sum_{-\infty}^{\infty} x_1(m)\, x_2(-m) = -1 + 0.5 = -0.5$$

$$x_3(1) = \sum x_1(m)\, x_2(1-m) = 1 + 1 + 1 = 3$$

$$x_3(2) = \sum x_1(m)\, x_2(2-m) = -2 + 1 + 2 + 1 = 2$$

$$x_3(3) = \sum x_1(m)\, x_2(3-m) = -0.75 + 2 - \cancel{2} - 2 = -2.75$$

$$x_3(4) = \sum x_1(m)\, x_2(4-m) = 0.75 + 4 + 2 = 6.75$$

$$x_3(5) = \sum x_1(m)\, x_2(5-m) = -2.5$$

$$x_3(6) = \sum_{-\infty}^{\infty} x_1(m)\, x_2(6-m) = -1.5$$

Q2.

1	1	2	3	2	2
1	1	2	3	2	2
4	4	2	5	1	1
1	1	2	6	3	3
2	2	4	6	7	7
2	2	4	6	7	7

low pass avg filter $\Rightarrow \frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

It. 1

$\frac{1}{9}(1+1+2+1+1+2+4+4+2)$
$=2$

$1/9\,(1+2+3+1+2+3+4+2+5)$
$=2.56$

$1/9\,(2+3+2+2+3+2+2+5+1)$
$=2.4$

$1/9\,(3+2+2+3+2+2+5+1+1)$
$=2.3$

It.2

$1/9\,(1+1+2+4+4+2+1+1+2)$
$=2$

$1/9\,(1+2+5+4+2+5+1+2+6)$
$=2.89$

$1/9\,(2+5+2+2+5+1+2+6+3$
$=2.89$

$1/9\,(5+2+2+5+1+1+6+3+3)$
$=2.89$

It 3

$1/9\,(4+4+2+1+1+2+2+2+4)$
$=2.4$

$1/9\,(4+2+5+1+2+6+2+4+6)$
$=3.56$

It 4

$1/9\,(1+1+2+2+2+4+2+2+4)$
$=2.2$

$1/9\,(1+2+6+2+4+6+2+4+6)$
$=3.67$

$= 3.56$

$\frac{1}{9}(2+5+1+2+6+3+4+6+7)$
$= 4$

$\frac{1}{9}(5+1+1+6+3+3+6+7+7)$
$= 4.33$

$\frac{1}{9}(2+6+3+4+6+7+4+6+7)$
$= 5$

$\frac{1}{9}(6+3+3+6+7+7+6+7+7)$
$= 5.78$

$$
\Rightarrow
\begin{matrix}
1 & 1 & 2 & 3 & 2 & 2 \\
1 & 2 & 3 & 2 & 2 & 2 \\
4 & 2 & 3 & 3 & 3 & 1 \\
1 & 2 & 4 & 4 & 4 & 3 \\
3 & 2 & 4 & 5 & 6 & 7 \\
2 & 2 & 4 & 6 & 7 & 7
\end{matrix}
$$

b)

$$\begin{matrix} 2 & 3 & 3 & 2 \\ 4 & 2 & 4 & 3 \\ 3 & 2 & 3 & 5 \\ 2 & 4 & 2 & 4 \end{matrix}$$

max val = 5

$2^{3} > 5$

8 levels

0 to 7

6 pix

$2 \rightarrow 3$.

Gray Lvl	0	1	2	3	4	5	6	7
No.of pix	0	0	6	5	4	1	0	0
Runn. sum	0	0	6	11	15	16	16	16
Run/Total	$\frac{0}{16}$	$\frac{0}{16}$	$\frac{6}{16}$	$\frac{11}{16}$	$\frac{15}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$
mx. x g Lvl.	0	0	2.6	4.8	6.5	7	7	7
	0	0	3	4	7	7	7	7

Gray	0	1	2	3	4	5	6	7
No pix	0	0	0	6	5	0	0	5

Q3.

$$\begin{matrix} 128 & 212 & 255 \\ 54 & 62 & 124 \\ 140 & 152 & 156 \end{matrix}$$

$2^{8} = 256$ ↳ L

$$s = c * \log(1+r)$$

Pixel Value	S		
128	$1 * \log(1+128)$	$= 2.11$	~ 2
212	$1 * \log(1+212)$	$= 2.32$	~ 2
255	$1 * \log(1+255)$	$= 2.4$	~ 2
54	$1 * \log(1+54)$	$= 1.74$	~ 2
62	$1 * \log(1+62)$	$= 1.79$	~ 2
124	$1 * \log(1+124)$	$= 2.09$	~ 2
140	$1 * \log(1+140)$	$= 2.14$	~ 2
152	$1 * \log(1+152)$	$= 2.18$	~ 2
156	$1 * \log(1+150)$	$= 2.17$	~ 2

o/p

$$\begin{matrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{matrix}$$

$$C = \frac{256}{\log(257)} = 10$$

Pixel Value	S			
128	$106 * \log(1+128)$	=	223.66	~ 224
212	$106 * \log(1+212)$	=	245.92	~ 246
255	$106 * \log(1+255)$	=	254.4	~ 254
54	$106 * \log(1+54)$	=	184.4	~ 184
62	$106 * \log(1+62)$	=	189.74	~ 190
124	$106 * \log(1+124)$	=	221.54	~ 222
140	$106 * \log(1+140)$	=	22684	~ 227
152	$106 * \log(1+152)$	=	231.08	~ 231
156	$106 * \log(1+150)$	=	230.02	~ 230

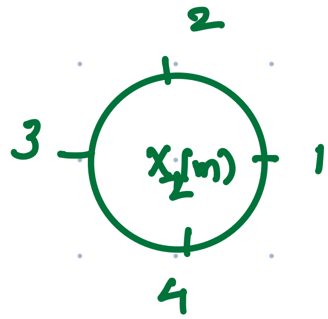
2025

Q1. Circular conv. & graph.

$x_1(n) = \{2, 1, 2, -1\}$ $\quad$ $x_2(n) = \{1, 2, 3, 4\}$

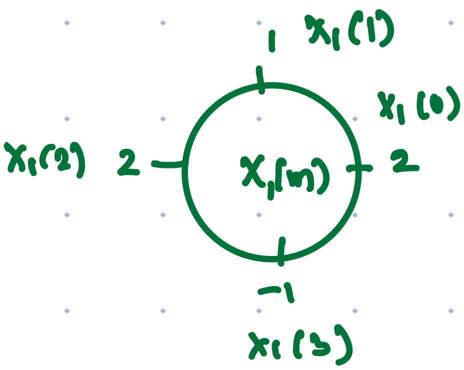
$x_1(m)$ $\quad$ $x_2(m)$ $\quad$ $0 \to 3$

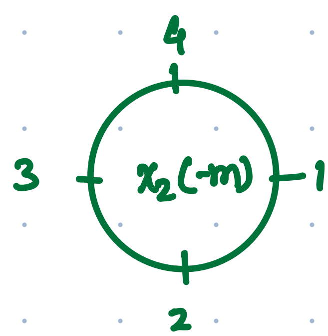
$$x_3(n) = \sum_{n=0}^{3} x_1(m) \cdot x_2((n-m))_4$$



$$x_3(0) = \sum_{n=0}^{3} x_1(m) \cdot x_2((-m))_4$$

↪ cw

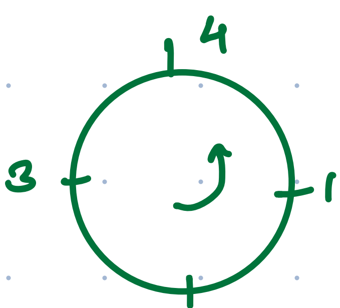




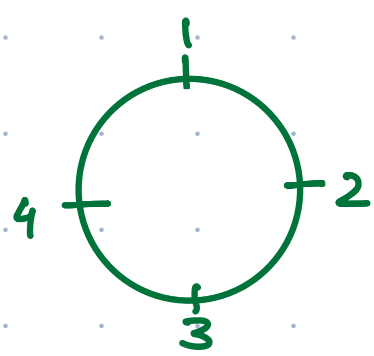
$$= \cancel{2} + 4 + 6 - \cancel{2} = 10$$

$$x_3(1) = \sum x_1(m)\, x_2(1-m)$$

↪ cw ↻ ACW



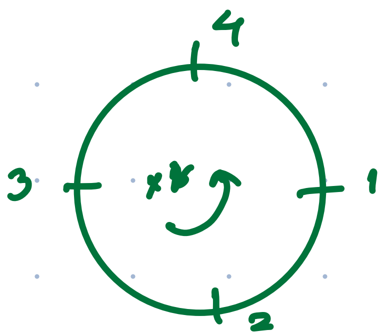
$\Rightarrow$

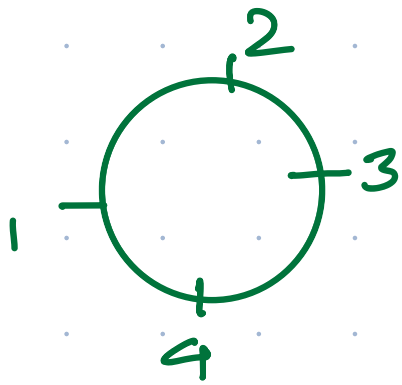


2

$= 4 + 1 + 8 - 3 = 10$

$x_3(2) = \sum x_1(m)\, x_2(2-m)$

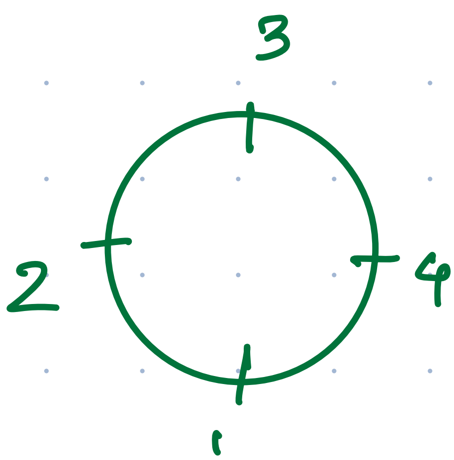




$= 6 + 2 + 2 - 4$

$= 6$

$x_3(3)$



$= 8 + 3 + 4 - 1$

$= 14$

Q2.

a) $h(n) = 0.2^n u(n)$ & stability

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

$$\sum_{n=0}^{\infty} (0.2)^n$$

$\hookrightarrow c < 1$

Infinite series

$$\frac{1}{1-0.2} < \infty$$

$\therefore$ stable

b) $\cos\left(\frac{5\pi}{9} n + 1\right)$

$\hookrightarrow \cos(\omega n \pm \theta)$

$\omega \cancel{n} = \frac{5\pi}{9} \cancel{n}$

$$\omega = 2\pi f$$

$$\frac{5\pi}{9} = 2\pi f$$

$$f = \frac{5}{18} \quad \frac{k}{N}$$ } rational ↳ periodic

$$\underline{\underline{N = 18}}$$

Q3.

gray lvl	0	1	2	3
	70	20	7	3
Run. sum	70	90	97	100
↳ / max	70/100	90/100	97/100	100/100
*3	2	3	3	3

0 → 2
1 → 3
2 → 3
3 → 3

g. lvl	0	1	2	3
no. of pix	0	0	70	30

↳ Lvl	0	1	2	3
no	0	0	70	30
Cum.	0	0	70	100
↳/total	0/100	0/100	70/100	100/100
*3	0	0	2	3

$2 \rightarrow 2$

$3 \rightarrow 3$

0	1	2	3
0	0	70	30

$\Rightarrow$ no change

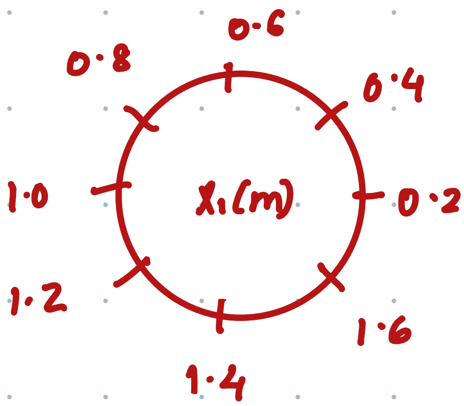
2024

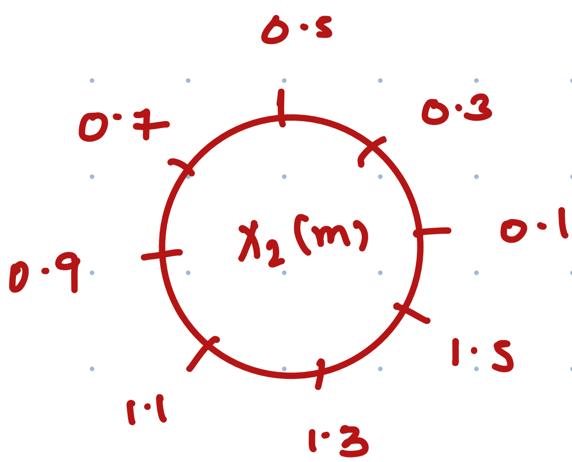
Q1. $x_1(n) = \{0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6\}$

$x_2(n) = \{0.1, 0.3, 0.5, 0.7, 0.9, 1.1, 1.3, 1.5\}$

Circular

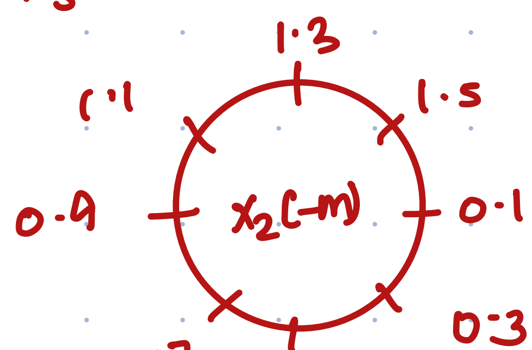
$$x_3(n) = \sum_{0}^{7} x_1(m) \cdot x_2((m-n))_8$$





CW.

$x_3(0) \Rightarrow \Sigma x_1(m) \cdot x_2(-m)$

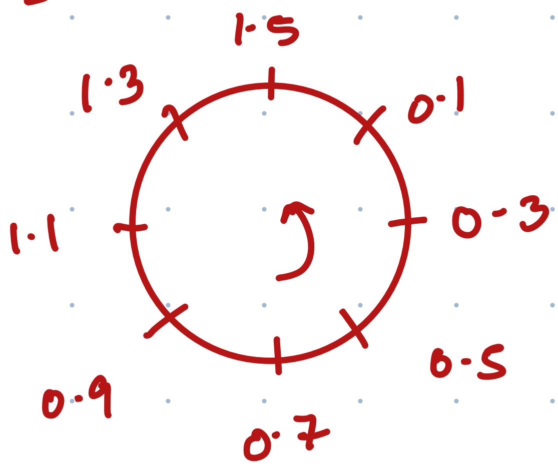


$\Rightarrow 0.02 + 0.6 + 0.78 + 0.88 + 0.9 + 0.84 + 0.7 + 0.48 = 5.2$

0·4

0·5

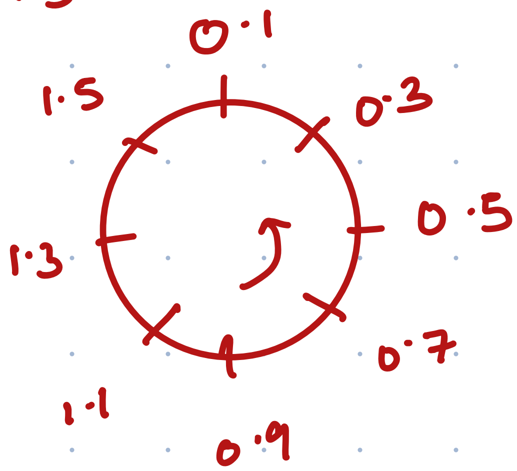
$x_3(1) \Rightarrow x_1(m)\, x_2(1-m)$



$\Rightarrow 0{\cdot}06 + 0{\cdot}04 + 0{\cdot}9 + 1{\cdot}04 + 1{\cdot}1 + 1{\cdot}08 + 0{\cdot}98 + 0{\cdot}7$

$= 6$

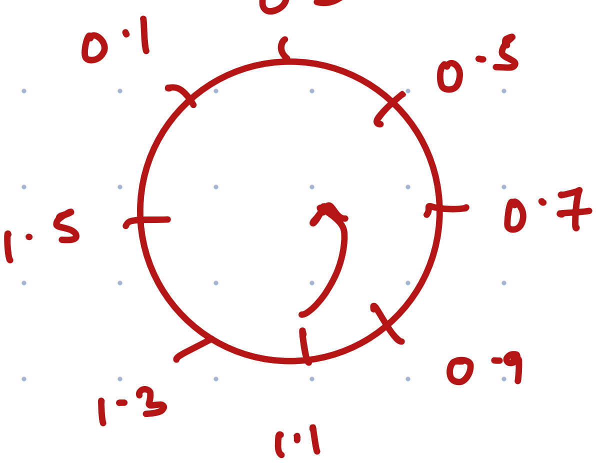
$x_3(2) \Rightarrow x_1(m) \cdot x_2(2-m)$



$= 0{\cdot}1 + 0{\cdot}12 + 0{\cdot}06 + 1{\cdot}2 + 1{\cdot}3 + 1{\cdot}32 + 1{\cdot}26 + 0{\cdot}98$

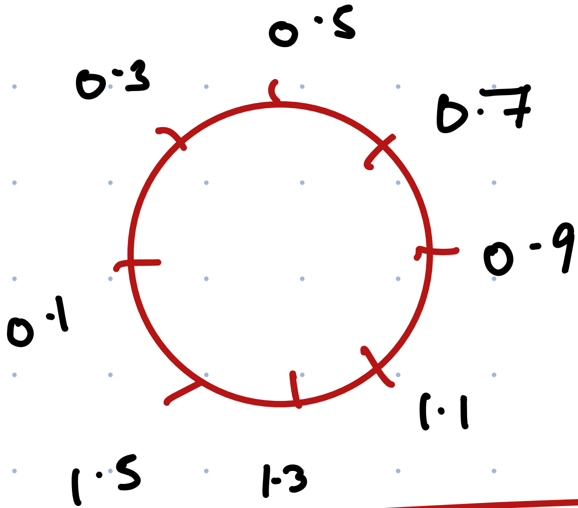
$= 6{\cdot}48$

$x_3(3) \Rightarrow x_1(m) \cdot x_2(3-m)$



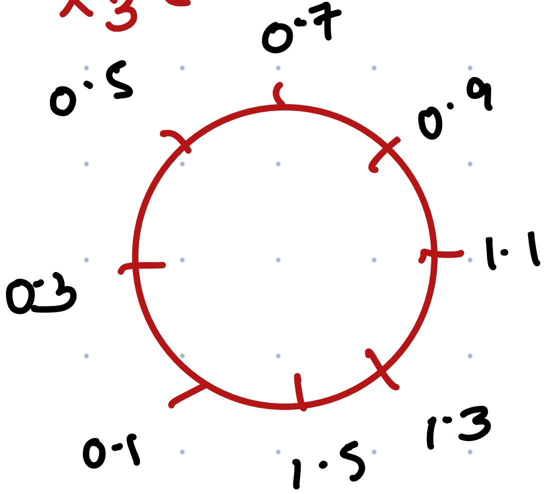
$= 6{\cdot}64$

$x_3(4)$



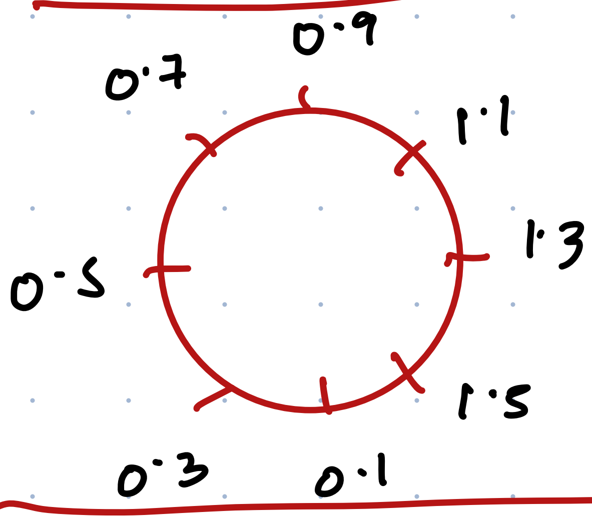
$= 6.48$

$x_3(5)$



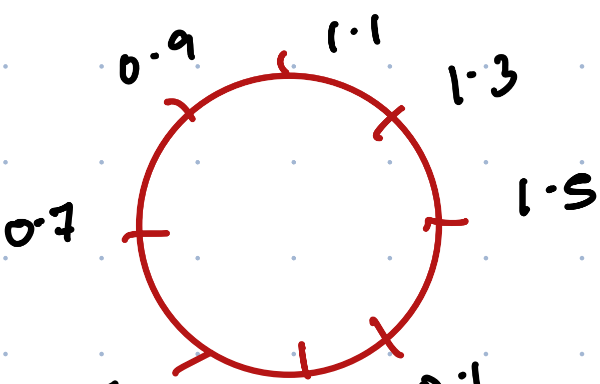
$= 6.0$

$x_3(6)$



$= 5.20$

$x_3(7)$



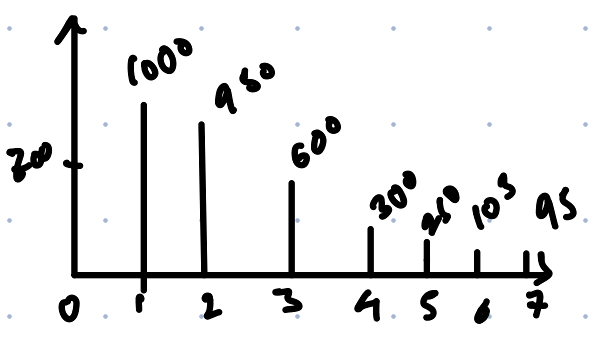
$= 4.08$

0·5 0·3

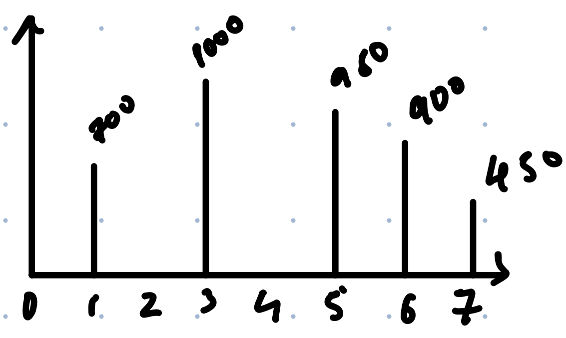
Q2.

gray lvl	0	1	2	3	4	5	6	7
No. of pix	700	1000	950	600	300	250	105	95
Run. sum	700	1700	2650	3250	3550	3800	3905	4000
/ total	$\frac{700}{4000}$	$\frac{1700}{4000}$	$\frac{2650}{4000}$	$\frac{3250}{4000}$	$\frac{3550}{4000}$	$\frac{3800}{4000}$	$\frac{3905}{4000}$	$\frac{4000}{4000}$
* 7	1	3	5	6	6	7	7	7

gry lvl	0	1	2	3	4	5	6	7
New no. of pix	0	700	0	1000	0	950	900	450



og



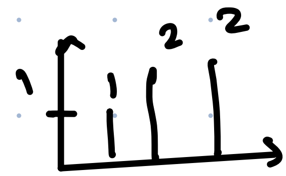
New

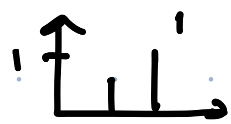
Q 2. $x(n) = \{1, \overset{\downarrow}{1}, 2, 2\}$

$y(n) = \{1, \underset{\uparrow}{0.5}, 1\}$

$$R_{xy}(n) = \sum_{k=-\infty}^{\infty} x(k)\, y(k-n)$$

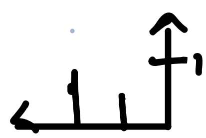
$n \rightarrow$	min	max
	$0-2$	$3-0$
	-2	3





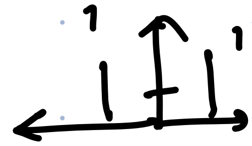
y(n)

$$R_{xy}(-2) = \sum_{-\infty}^{\infty} x(k)\, h(k-n)$$



$$= \sum x(k)\, h(k+2)$$

$$= 1$$



$$R_{xy}(-1) = \sum x(k)\, h(k+1)$$

$$= 0.5 + 1 = 1.5$$

<u>2022</u>

$$R_{xy}(0) = \sum x(k)\, h(0)$$

Q1

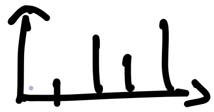
$= 1 + 0.5 + 2$

a) $y(n) = \sin\left(\frac{2\pi}{7} n^2\right)$

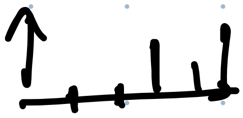
$= 3.5$

$R_{xy}(1) = x(n+1) = x(n+N)$

$R_{xy}(2) = 2 + 1 = 3$



$R_{xy}(3) = x(n+N) \Rightarrow$



$$\sin\left[\frac{2\pi}{7}(n+N)^2\right]$$

$$\sin\left[\frac{2\pi}{7}(n^2+N^2+2nN)\right]$$

$$\sin\left(\underbrace{\frac{2\pi}{7}n^2}_{\text{og}} + \underbrace{\frac{2\pi}{7}N^2 + \frac{2\pi}{7}nN}_{\text{extra}}\right)$$

$\frac{\cancel{2\pi}}{7} N^2 = \cancel{2\pi} f$ $\qquad$ $\frac{\cancel{2\pi}}{7} N = \cancel{2\pi} f$

$N = \sqrt{7f}$ $\qquad$ $N = 7f$

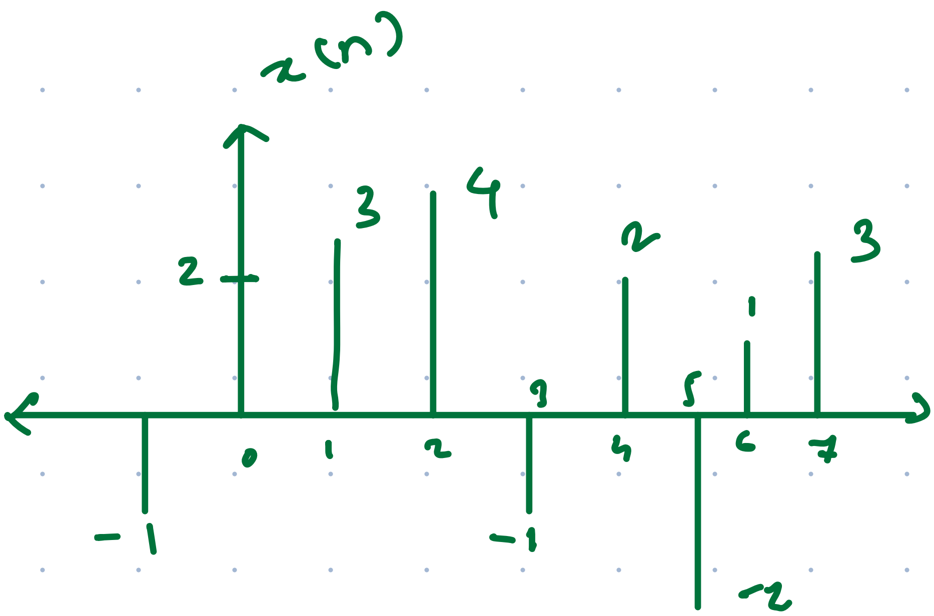
$N = \sqrt{77}$ $\qquad\qquad$ $N = 77$

$= \sqrt{7 \times 7}$ $\qquad\qquad$ 7, 14, 21, 28, 35,

$= 7$ $\qquad\qquad$ 42, 49

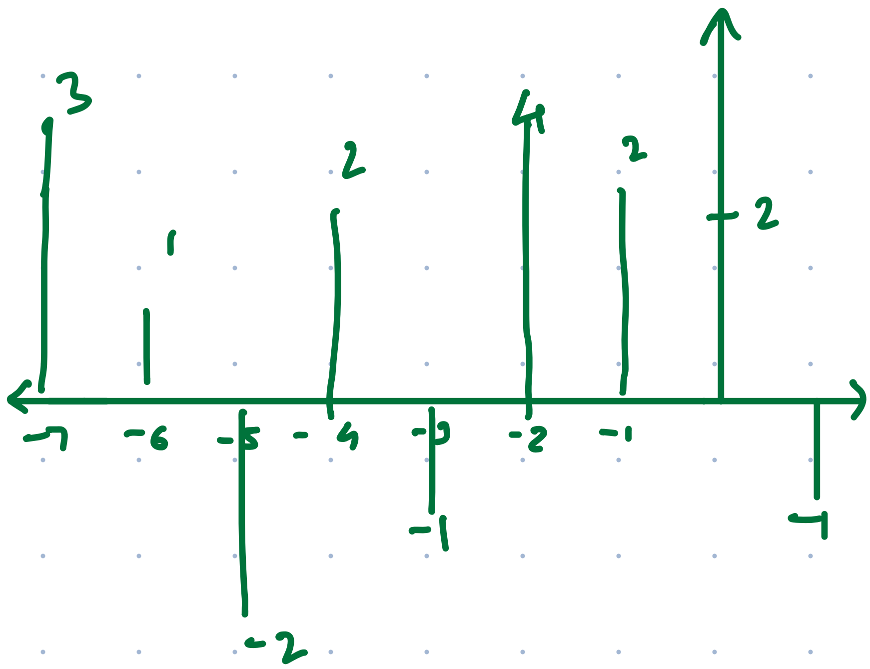
$N = 7$

b) $\{-1, \underset{\uparrow}{2}, 3, 4, -1, 2, -2, 1, 3\}$



$$x_e(n) = \frac{x(n) + x(-n)}{2}$$

$$x_o(n) = \frac{x(n) - x(-n)}{2}$$



Q2. Time inv.

$\Rightarrow$

a) $y(n) = x(n) + x(n+1)$

$\hookrightarrow$ $y_1(n) = x(n-m) + x(n-m+1)$

$y_2(n) = x(n-m) + x(n-m+1)$ $\Big\} =$

$\therefore$ inv.

b) $y(n) = 4n\, x(n)$

$y_1(n) = 4n\, x(n-m)$

$$y_2(n) = 4(n-m)\,x(n-m)$$ $\neq$ variant

$\Rightarrow$ Linear

a) $y(n) = 2n\,x(n)$

(i) $x_1(n) \longrightarrow \boxed{H} \longrightarrow y_1(n)$ $\qquad H\{x_1(n)\} = 2n\,x_1(n)$

$x_2(n) \longrightarrow \boxed{H} \longrightarrow y_2(n)$ $\qquad H\{x_2(n)\} = 2n\,x_2(n)$

$$a_1 y_1(n) + a_2 y_2(n) = a_1 2n\,x_1(n) + a_2 2n\,x_2(n)$$

(ii) Consider combo.

$x_3(n) = a_1 x_1(n) + a_2 x_2(n)$

$\longrightarrow \boxed{H} \longrightarrow y_3(n)$ $\qquad H\{x_3(n)\} = 2n\,x_3(n)$

$$= 2n\left[a_1 x_1(n) + a_2 x_2(n)\right]$$

$\hookrightarrow$ linear

b) $y(n) = e^{x(n)}$

$x_1(n) \longrightarrow \boxed{H} \longrightarrow y_1(n)$ $\Rightarrow H\{x_1(n)\} = e^{x_1(n)}$

$x_2(n) \longrightarrow \boxed{H} \longrightarrow y_2(n)$ $= e^{x_2(n)}$

$$a_1 y_1(n) + a_2 y_2(n) = a_1 e^{x_1(n)} + a_2 e^{x_2(n)}$$

comb. of linear

$x_3(n) = a_1 x_1(n) + a_2 x_2(n)$

$x_3(n) \longrightarrow \boxed{H} \longrightarrow$ $y_3(n) = e^{x_3(n)}$

$\hookrightarrow e^{a_1 x_1(n) + a_2 x_2(n)}$

Q- 3 bit $2^3 = 8$ levels 0-7

a) $r_1 = 3$ $r_2 = 5$

$r_1 \leq r \leq r_2$

a
with bg

$L-1 = 7$ $3 \leq r \leq 5$

0

w/o bg

$L-1 = 7$ $3 \leq r \leq 5$

r

2	4	5	7
6	5	7	0
1	3	0	4

with bg

0	7	7	0
0	7	0	0
0	7	0	7

w/o bg

2	7	7	7
6	7	7	0
1	7	0	4

b) M L

010 100 101 111

110 101 111 000

001 011 000 100

LSB				Mid				MSB			
0	0	1	1	1	0	0	1	0	1	1	1
0	1	1	0	1	0	1	0	1	1	1	0
1	1	0	0	0	1	0	0	0	0	0	1